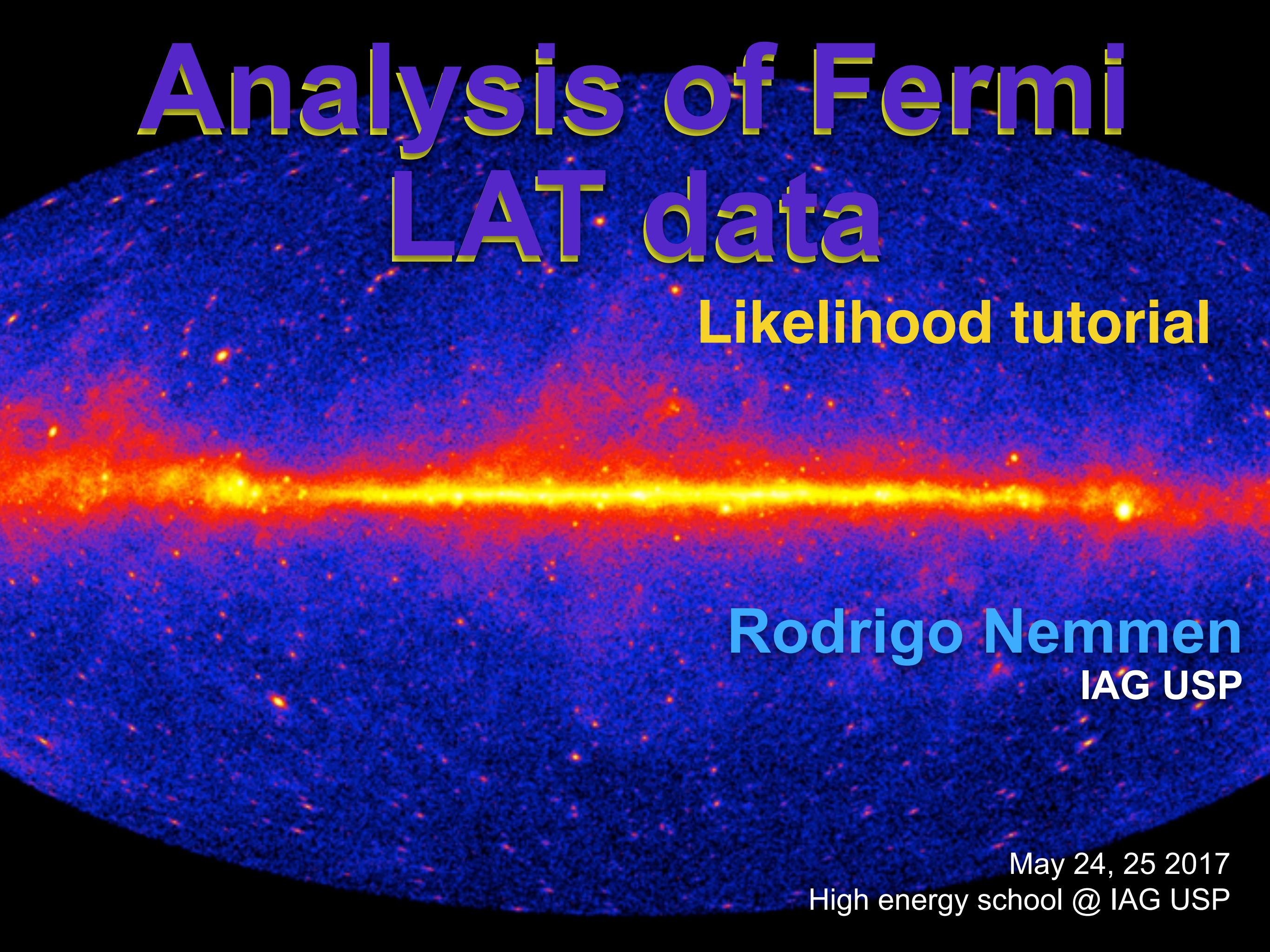


The background of the slide is a Fermi Large Area Telescope (LAT) gamma-ray sky map. It shows a wide field of view with a prominent, bright, horizontal band of high-energy emission representing the Milky Way galaxy. The rest of the sky is filled with numerous smaller, point-like sources of gamma-ray emission, appearing as red and orange dots against a dark blue background.

Analysis of Fermi LAT data

Likelihood tutorial

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High energy school @ IAG USP

Overview of activities

Day 1

- ◆ Introduction, overview
- ◆ Obtaining and exploring LAT data
- ◆ Inspecting the data: count map



Day 2

- ◆ **Basics of modeling**
- ◆ Likelihood analysis of a blazar
- ◆ Create a SED
- ◆ Produce a light curve

Structure of this talk

- **Basic theory of likelihood analysis**

see jupiter
notebook

- Likelihood fit → Characterize spectra of a source

- obtain spectral energy distribution (SED)

- build a light curve

hands-on

Fundamental questions of science

statistics

Model fitting: given this model, what parameters best fit my data?

Model selection: given two potential models, which better describes my data?

Model fitting

given this model, what
parameters best fit my data?

Maximum likelihood approach

Frequentist

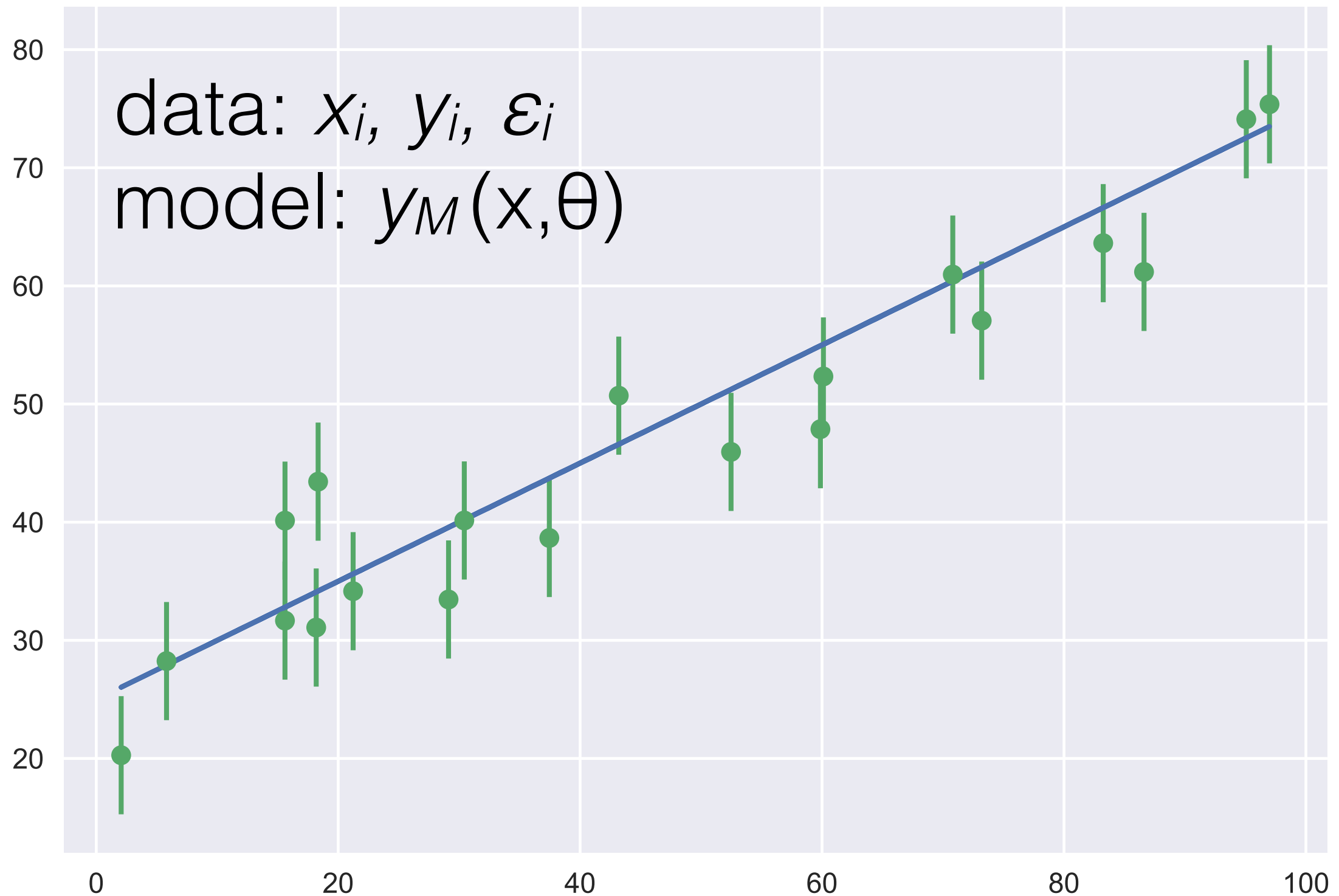
$$P(\textit{data} \mid \textit{scientific model})$$

$$P(D \mid \theta)$$

likelihood

$$\frac{\partial \mathcal{L}}{\partial \theta} = 0$$

How to write the likelihood function for the problem below?



Probability for a single measurement

$$P_i = \frac{1}{\sqrt{2\pi\epsilon_i^2}} e^{-\frac{(y_i - y_M)^2}{2\epsilon_i^2}}$$

Likelihood: $\mathcal{L} = \prod_i P_i$

Log-likelihood:

$$\log \mathcal{L} = -\frac{1}{2} \sum_{i=1}^N \left\{ \log(2\pi\epsilon_i^2) + \frac{[y_i - y_M(x_i; \theta)]^2}{\epsilon_i^2} \right\}$$

**What about *Fermi*
LAT data?**

Experiment counting events during a fixed interval:

Poisson distribution

n: number of detected events

m: expected number (model)

Probability of
observing **n**
events

$$P = \frac{m^n e^{-m}}{n!}$$

Standard
deviation $= \sqrt{m}$

Experiment counting events during a fixed interval:

Poisson distribution

$$P = \frac{m^n e^{-m}}{n!}$$

Standard
deviation $= \sqrt{m}$

Experiment counting events during a fixed interval: Poisson distribution

Detected: $n = 3$

Expected: $m = 2$

Noise: $\sqrt{m} = 1.4$

$$P = \frac{m^n e^{-m}}{n!}$$

Standard
deviation $= \sqrt{m}$

γ

γ

γ

e^-

e^+

e^-

e^+

e^-

e^+

Likelihood for Poisson distribution

Binned likelihood

$$\mathcal{L} = \prod_i e^{-m_i} \prod_i \frac{m_i^{n_i}}{n_i!} = e^{-N_{\text{pred}}} \prod_i \frac{m_i^{n_i}}{n_i!}$$

For bin sizes
infinitesimally small, $n_i = 1$

$$\mathcal{L} = e^{-N_{\text{pred}}} \prod_i m_i$$

Unbinned likelihood

Log-likelihood

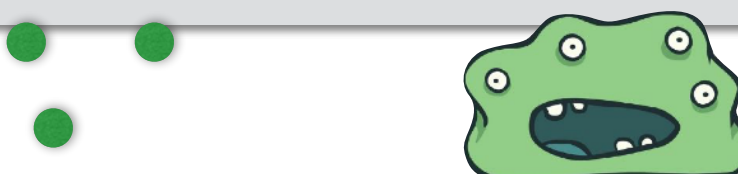
$$\log \mathcal{L} = \sum_i \log m_i - N_{\text{pred}}$$

We need to maximize this likelihood function numerically

$$\log \mathcal{L} = \sum_i \log m_i - N_{\text{pred}}$$

total predicted
number of
counts

$$\int_{\text{ROI}} \left(\text{instrument response function} \right) \left(\text{source model} \right) dE d\mathbf{p} dt$$

$$\text{source model} = \text{point sources} + \text{diffuse sources} + \text{other stuff}$$


Model selection

given two potential models, which
better describes my data?

Is there a source in a given position of the sky?

$$TS = -2 \log \left(\frac{\mathcal{L}_{\text{max},0}}{\mathcal{L}_{\text{max},1}} \right)$$

maximum likelihood: model without an additional source (the “null hypothesis”)

maximum likelihood: model with additional source

Rule of thumb:

$$TS = 100 \Rightarrow \approx 10\sigma$$

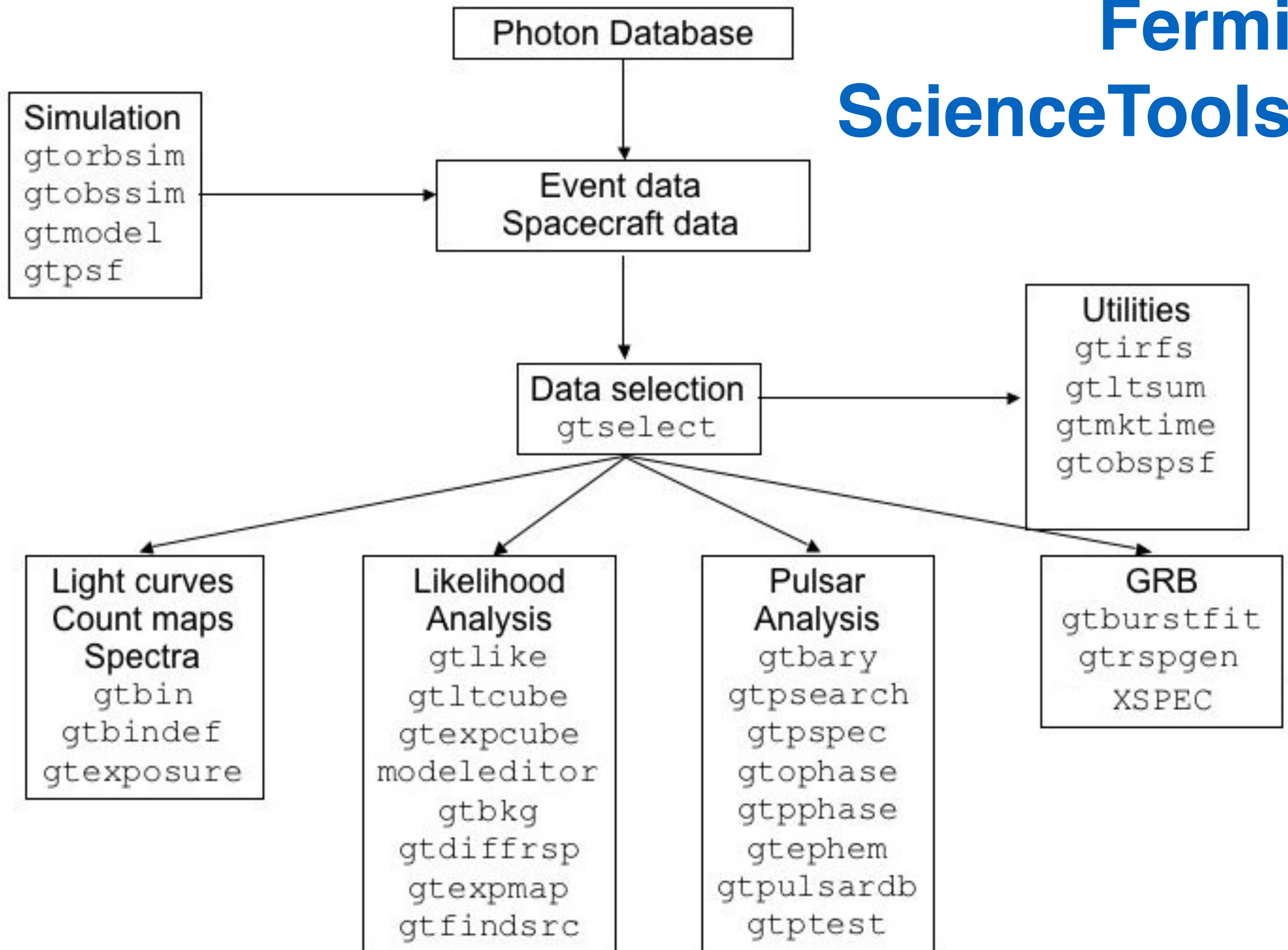
$$TS = 25 \Rightarrow \approx 5\sigma$$

Hands-on activity

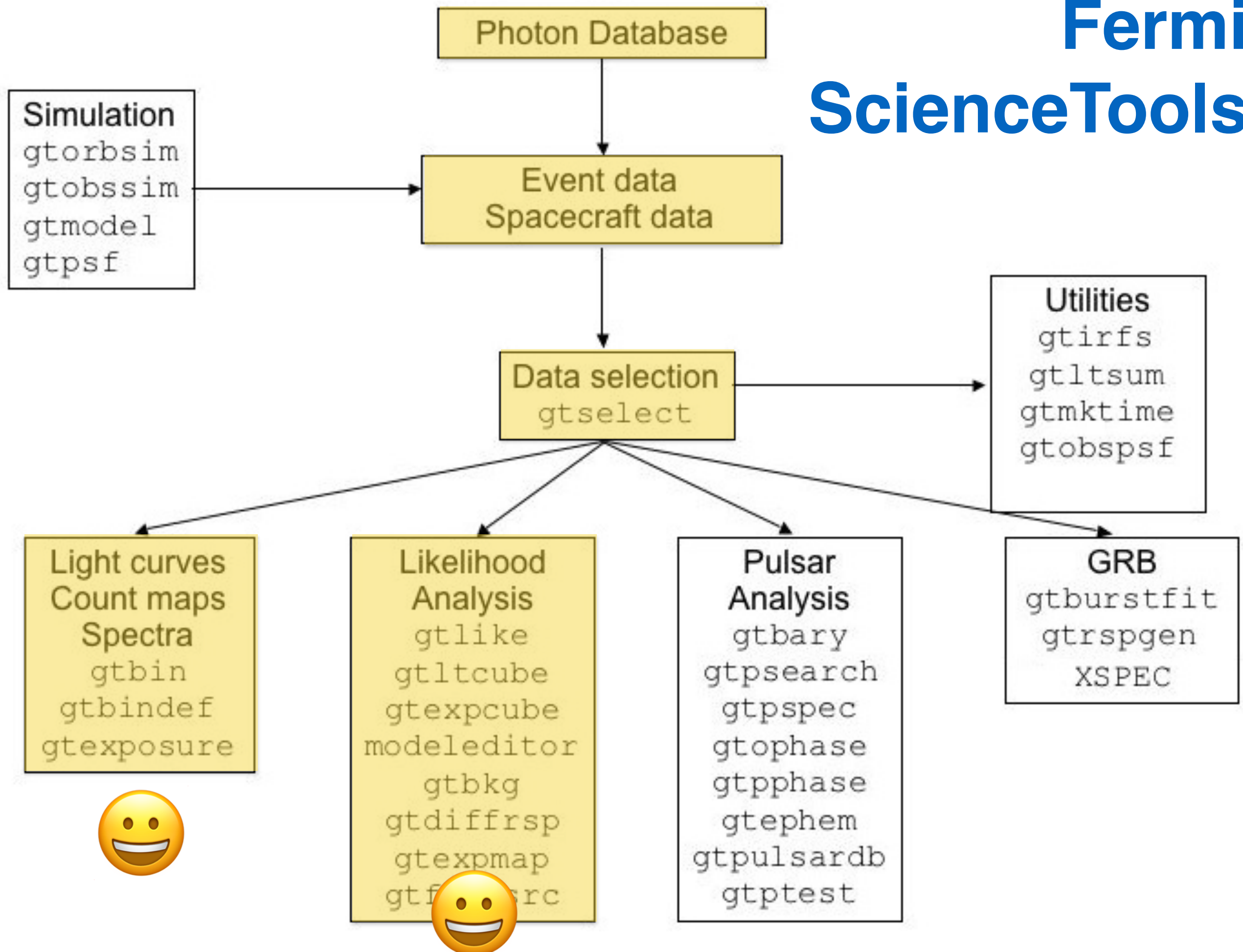
Steps for modeling of *Fermi* LAT data

- 1 Define your source model
- 2 Select data + ROI cuts `gtselect`
- 3 Select good time intervals (GTI) `gtmktime`
- 4 Bin data, create a count map `gtbin`
- 5 Compute useful quantities: live time cube, binned exposure cube `gtltcube`, `gtexpcube2`
- 6 Maximize likelihood numerically, get initial estimate of parameters `gtlike`
- 7 Optimize fit: improve parameter estimate

Fermi ScienceTools



Fermi ScienceTools



Python wrappers make analysis of Fermi data much easier: Enrico and fermipy

We will use Enrico:

<http://enrico.readthedocs.io/en/latest/>

Tutorial website

<https://github.com/rsnemmen/Fermi-LAT-tutorial>

Tutorials

Day one

- 4:30-5:00: [Introduction, overview of activities and tools \(slides\)](#)
- 5:00-5:30: [Obtaining and preparing LAT data for your favorite source](#)
- 5:30-6:30: [Exploring LAT data: Plotting the counts map](#)

Day two

- 4:30-5:00: Overview of activity, basic theory of spectral modeling: [slides](#), [jupyter notebook](#)
- 5:00-5:30: [Getting a flux: Likelihood analysis](#) **1**
- 5:30-6:30: [Creating a spectrum \(SED\)](#) **2**
- Bonus: [Producing a light-curve](#)

start here

3
**come
here
later**



**Please do not download large files
during the tutorial or the WIFI network
will *overload***

**We will distribute the software and data
you need via USB sticks**





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